Lesson notes · for you

Fair for everyone

AI · Level 3 · Lesson 4

Time
25 min

Materials
A phone with voice typing (yours, or a family member's with permission); Three short sentences written on paper, in each language you will test; A notebook for your results table

Learning goals

- Find who a model fails by testing each group separately, not only overall.
- Fix bias at its source: collect the missing examples, then test again.
- Explain why a fair model matters to the people it judges.

Activity: Fairness audit

1. Choose (3 min): Pick voice typing you are allowed to use (or a translation app's microphone, checking only the words it writes) and write three short, everyday sentences. If a volunteer speaks another language, write the sentences in that language too.
2. Ask (4 min): Find at least four volunteers who differ in age, accent or language. Tell them what the test is for and that no recordings will be kept.
3. Test (8 min): Each person says the same three sentences once. Count the words the tool writes correctly, out of all the words, and write down the share (such as 10 out of 12), not the person's name.
4. Table (5 min): Make a table (Person A, Person B…) with each person's share, and add the overall share at the bottom.
5. Explain (5 min): Does the overall share hide anyone? Write who the tool understood least, a likely gap in its training data, and how its makers could fix it. With one person for each kind of voice, call it a clue, not proof.

Works in a small group or at home with family. Only test people who agree, keep no recordings, and use letters instead of names.

Discussion

- Why can a model with a great overall score still be unfair?
- Who might be missing from the data behind a tool you use every day?
- Is it ever fair to ask people to change how they speak or write so a model understands them?

Check yourself

- One group a model I use might fail, and how I would test it.

Try it this week

Ask family members first, then test voice typing with them (or the microphone in a translation app, checking only the words it writes down): young and old, different accents or languages. Have each person say the same three sentences, count the words it writes correctly out of all the words, and compare the shares, such as 10 out of 12. Write down who it understands best and why that might be. With one person for each kind of voice, your result is a clue, not proof.

Worksheet

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Name _____ Class _____ Date _____

1 **Your voice model struggles with older people's voices. What's the best fix?**

- ◯ Ask older people to speak differently
- ◯ Report only the overall score
- ◯ Record more older voices, with permission

2 **A leaf spotter was trained mostly on photos from a sunny valley. Which steps would show whether it is fair to farmers in the cloudy hills? Choose all that apply.**

- ☐ Check only its overall score
- ☐ Test it on photos from the hills
- ☐ Test it again on its training photos
- ☐ Compare the hills' score with the valley's
- ☐ Count the sick leaves it misses in each region

3 **Put the steps for fixing a biased model in order.**

- ☐ Train the model again
- ☐ Collect and label examples from that group, with permission
- ☐ Test every group again
- ☐ Test each group separately
- ☐ Find the group with the lower score

4 **Complete the sentence.**

Bias can hide inside the _____ score, so test each _____ on its own, and close the gap by collecting the _____ examples.

Word bank: group · overall · perfect · missing · training

✎ **Think of an AI tool you or your family use. Which group of people might it work less well for, and how could you find out?**

Check your answers

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1. **Record more older voices, with permission**

   The gap comes from the data, so fix it there: collect the missing voices with permission, retrain and test every group again.

2. **Test it on photos from the hills; Count the sick leaves it misses in each region; Compare the hills' score with the valley's**

   Fairness shows up group by group: test on hill photos, count the misses in each region and compare.

3. **Top to bottom: 4, 3, 5, 1, 2**

   You can't fix a gap you haven't found, so testing by group comes first.

4. **Bias can hide inside the overall score, so test each group on its own, and close the gap by collecting the missing examples.**

   One overall number can hide a group the model fails.